

Q1. What is a Calculated Field and why is it important in business dashboards?

Ans: A **calculated field in Tableau** is a custom field that you create using formulas to transform or derive new data from your existing dataset. Instead of relying only on raw data columns, you can define your own logic—such as mathematical calculations, conditional statements, string manipulation, or date functions—and use the result as a new field in your analysis.

Calculated fields are essential because they make dashboards more meaningful and actionable:

1. Custom metrics tailored to business needs

Businesses rarely rely only on raw data. Calculated fields let you define KPIs like margins, growth rates, or customer segments that reflect real business performance.

2. Better insights without altering source data

You don't need to modify the original dataset. Calculations happen within Tableau, which keeps the data source clean while enabling flexible analysis.

3. Dynamic and interactive analysis

Calculated fields update automatically when filters or parameters change, making dashboards responsive and interactive.

4. Simplifies complex logic

You can embed business rules (like discount tiers or performance categories) directly into the dashboard, making it easier for users to interpret results.

5. Improves decision-making

By turning raw numbers into meaningful insights, calculated fields help stakeholders quickly understand trends and take informed actions.

Q2. Why can't parameters change visuals directly in Tableau?

1. Parameters are independent of the data model

Unlike filters or fields, parameters are not tied to any specific dataset. They simply store a value (number, string, date), so Tableau has no built-in instruction on how that value should affect a chart.

2. No automatic action logic

A visualization changes only when Tableau knows *what to do*. Filters remove data, dimensions slice it, and measures aggregate it. Parameters don't do any of that unless you explicitly connect them.

3. Require calculated fields or actions

To make a parameter influence a visual, you must use it inside:

- a **calculated field** (e.g., dynamic measure selection, conditional logic)
- a **parameter action** (to update values based on user interaction)

Without this link, the parameter just sits there unchanged.

Q3. When should EXCLUDE LOD be used?

EXCLUDE Level of Detail (LOD) expression is used when you want to remove one or more dimensions from the level of aggregation in your view, while still keeping those dimensions visible in the chart.

When you should use EXCLUDE LOD

1. To compare a detailed value with a higher-level total

Use EXCLUDE when your view is more granular, but you want a calculation at a broader level.

Example:

You're viewing **Sales by Region and Category**, but you want **total Sales per Region (ignoring Category)** for comparison.

→ EXCLUDE Category will give you the region-level total even though Category is still in the view.

2. To calculate percentages of totals

EXCLUDE is ideal for “part-to-whole” analysis.

Example:

- Show Category-wise sales
- Calculate % contribution to total Region sales
→ EXCLUDE Category helps compute the denominator (total Region sales).

3. To remove extra granularity added to the view

Sometimes adding a dimension (like Sub-Category) splits your numbers more than you want.

→ EXCLUDE lets you “ignore” that extra detail for specific calculations without removing it from the visualization.

4. When FIXED is too rigid

A FIXED LOD ignores filters (unless they're context filters), which can be limiting.

EXCLUDE respects most filters in the view, making it more dynamic for interactive dashboards.

Q4. Difference between FIXED and INCLUDE LOD expressions?

In Tableau, **FIXED** and **INCLUDE** LOD expressions both control the level at which calculations happen—but they behave quite differently in terms of **granularity and interaction with the view and filters**.

Core difference

- **FIXED** → Ignores the view's level of detail and computes at a **specified, fixed level**
- **INCLUDE** → Adds extra granularity to the view and computes at a **more detailed level than what's currently shown**

FIXED LOD

Definition:

Calculates a value at a specific level of detail, regardless of what dimensions are in the view.

Example:

```
{ FIXED [Region] : SUM([Sales]) }
```

This always returns total sales per Region, even if your view includes Category or Sub-Category.

Key traits:

- Independent of the view (dimensions in the chart don't affect it)
- Ignores most filters (except **context filters**)
- Useful for stable KPIs (e.g., total sales, total customers)

Use when:

- You want a consistent value no matter how the view changes
- You need totals, benchmarks, or reference values

INCLUDE LOD

Definition:

Adds dimensions to the calculation, making it more granular than the current view, and then aggregates back up.

Example:

```
{ INCLUDE [Customer Name] : SUM([Sales]) }
```

If your view is at Region level, this calculates sales per Customer first, then aggregates to Region.

Key traits:

- Depends on the view (it builds on existing dimensions)
- Respects filters in the view
- Great for “average of detailed values” type calculations

Use when:

- You need calculations at a finer level than what's displayed
- You're doing things like:
 - Average sales per customer

- Average order value

Simple analogy

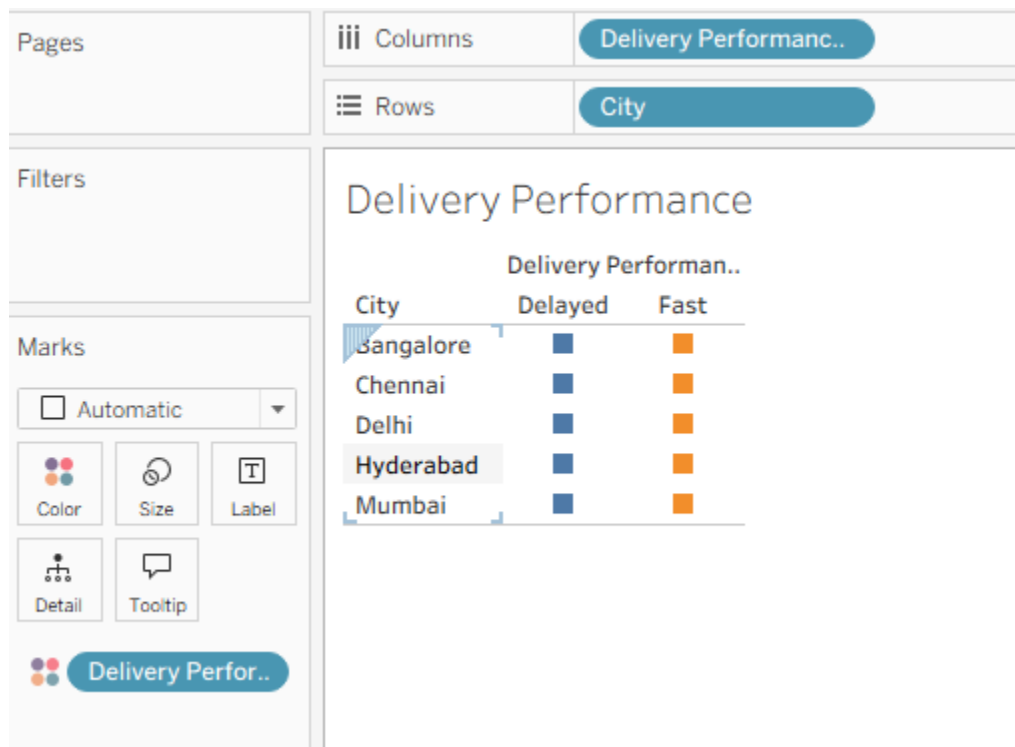
- **FIXED** = “Ignore the view, calculate exactly at this level.”
- **INCLUDE** = “Go deeper than the view, then roll it back up.”

Q5. Task 1: Delivery Performance Classification (Calculated Field)

Business Need: Operations want to track delivery efficiency.

Task Classify orders as Fast or Delayed.

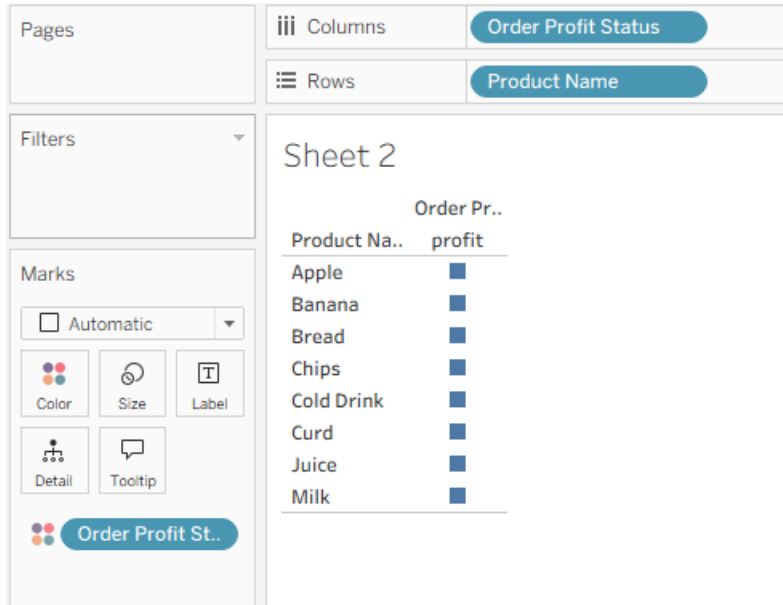
Condition $\text{Delivery} \leq 30 \text{ min} \rightarrow \text{Fast}$ Else $\rightarrow \text{Delayed}$



Task 2: Loss-Making Orders Identification

Business Need High sales do not always mean high profit.

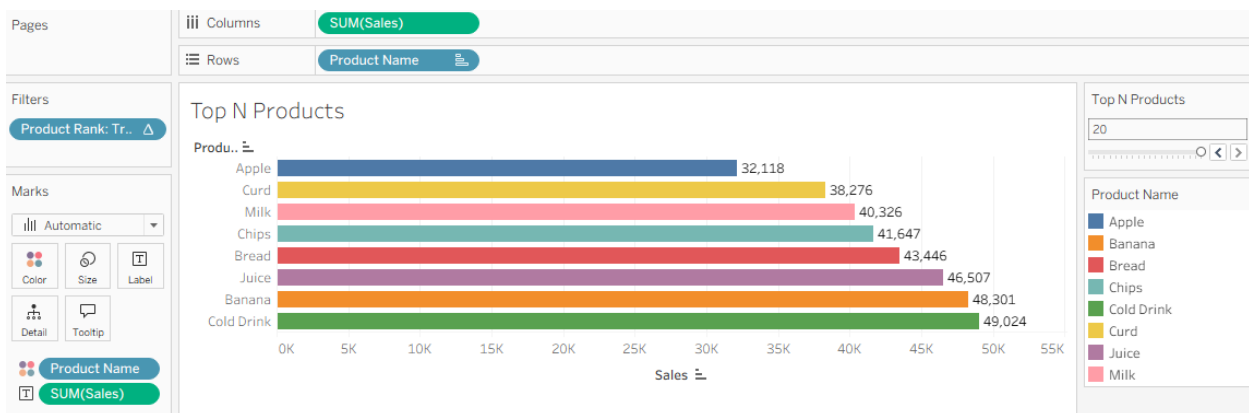
Task Identify orders resulting in loss.



Task 3: Top N Products Analysis (Parameter)

Business Need The product team wants to focus on top-performing products.

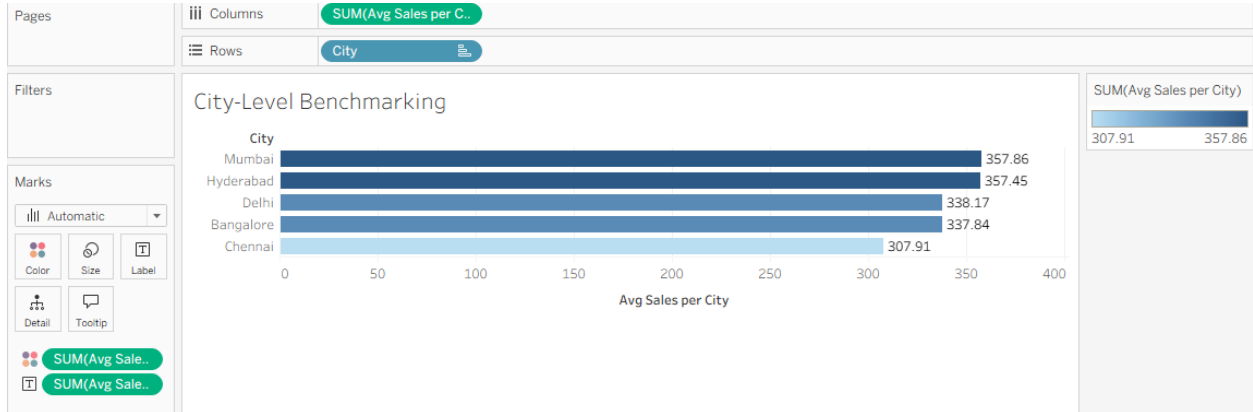
Task Create a Top N Products dynamic analysis.



Task 4: City-Level Benchmarking (FIXED LOD)

Business Need Compare city performance independent of category filters.

Task Calculate average sales per city using FIXED LOD.



Task 5: Customer/Product Detail Impact (INCLUDE LOD)

Business Need The sales team wants average sales considering product-level detail, even when product is not shown.

Task Create a calculation that finds the average sales at the product level, but allows the result to be used at a higher level (such as Category or Region). Use INCLUDE LOD.

